

NETWORK ADMINISTRATION USING CISCO (315)

STATE – 2023

TOTAL POINTS

_____ ***(100 points)***

Failure to adhere to any of the following rules will result in disqualification:

- 1. Contestant must hand in this test booklet and all printouts. Failure to do so will result in disqualification.**
- 2. No equipment, supplies, or materials other than those specified for this event are allowed in the testing area. No previous BPA tests and/or sample tests or facsimile (handwritten, photocopied, or keyed) are allowed in the testing area.**
- 3. Electronic devices will be monitored according to ACT standards.**

No more than sixty (60) minutes testing time

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Choose the *best* answer to the following questions. Mark answers clearly if using a scantron form. Mark A if the statement is true. Mark B if the statement is false.

1. What is a function of a switch?
 - a. It connects multiple IP networks
 - b. It controls the flow of data via the use of Layer 2 addresses
 - c. It provides segmentation at Layer 2
 - d. It builds a routing table based on ARP requests
2. Which one of the following IP address settings is not needed for local communication?
 - a. IP Address
 - b. Subnet Mask
 - c. DNS Server
 - d. Gateway
3. In a complete IPV6 address, how many groups of hexadecimal digits are divided by colons?
 - a. 4
 - b. 2
 - c. 6
 - d. 8
4. Which of the following protocols is considered a layer-5 protocol?
 - a. IP
 - b. TCP
 - c. NFS
 - d. UDP
5. Telnet allows a user to connect securely to a remote device and ensures all communication is encrypted
 - a. True
 - b. False
6. Only TFTP allows safe file transfer functionalities
 - a. True
 - b. False
7. What device divides broadcast domains?
 - a. Bridge
 - b. Switch
 - c. Router
 - d. Repeater
8. Which of the following port modes commonly describes a port that allows a single VLANs to go through?
 - a. Access Port
 - b. Trunk Port
 - c. Secure Port
 - d. General Port
9. Systems on different VLANs able to communicate between each other by using a Router
 - a. True
 - b. False

10. Which device determines the path that messages should take through internetworks?
 - a. Guider
 - b. Firewall
 - c. Web server
 - d. Router

11. Which of the following port modes describes a port that allows a group of VLANs to go through?
 - a. Access Port
 - b. Trunk Port
 - c. Secure Port
 - d. General Port

12. Which field in the IPv4 header is used to verify that there is no error on the packet?
 - a. Time-to-Live
 - b. Sequence Number
 - c. Acknowledgment Number
 - d. Checksum

13. What is one advantage that IPv6 offers over IPv4?
 - a. Smaller-sized header
 - b. Little requirement for processing checksums
 - c. Built-in QoS
 - d. Supports VLSM

14. What is the next step in the router boot sequence after the Bootstrap runs the program POST?
 - a. Tries to load the IOS from flash
 - b. Locate and load the startup-config file from NVRAM
 - c. Load the bootstrap program from ROM
 - d. Load the running-config file from RAM

15. What command is used to display the IP addresses assigned on the switch/router?
 - a. Show ip interface brief
 - b. Show ip
 - c. Show addresses
 - d. Show interface status

16. What does the IP address 192.168.4.159/27 represent?
 - a. Network address
 - b. Multicast address
 - c. Broadcast address
 - d. Host address

17. What type of address is automatically assigned to an interface when IPv6 is enabled on that interface?
 - a. Global unicast
 - b. Link-local
 - c. Loopback
 - d. Unique local

18. What is a reason a network administrator might want to create subnets?
 - a. Simplifies network design
 - b. Allow more networks to be made
 - c. Reduction in number of routers needed
 - d. Reduction in number of switches needed

19. What happens if two or more switches are connected through a router?
 - a. The number of broadcast domains is increased
 - b. The size of the broadcast domain is increased
 - c. The number of collision domains is reduced
 - d. The size of the collision domain is increased

20. How many assignable host addresses are available on the network 192.168.87.0 with a subnet mask of 255.255.255.224?
 - a. 34
 - b. 28
 - c. 30
 - d. 32

21. Which feature allows different communication streams to occur at the same time without a single stream using the entire bandwidth?
 - a. Window size
 - b. Multiplexing
 - c. Port numbers
 - d. Acknowledgments

22. What type of information is included in the network header?
 - a. Destination and source logical addresses
 - b. Destination and source physical addresses
 - c. Destination and source port numbers
 - d. Encoded application data

23. What part of the header is used by TCP to reorder segments?
 - a. Checksum
 - b. ACK numbers
 - c. Port numbers
 - d. Sequence numbers

24. Which application layer protocol is used natively to provide file-sharing services to Linux applications?
 - a. HTTP
 - b. NFS
 - c. DHCP
 - d. SMB

25. Which type of network threat includes an intruder engaging with the system in order to find out about vulnerabilities?
 - a. DoS attacks
 - b. Access attacks
 - c. Reconnaissance attacks
 - d. Trust exploitation

26. If the router is using an information base and an adjacency table, it means it's using which packet-forwarding method?
- Fast switching
 - Cisco Express Forwarding
 - Process switching
 - Flow process
27. What route would have the highest administrative distance?
- Internal BGP route
 - Static route
 - Route received through the EIGRP routing protocol
 - Route received through the OSPF routing protocol
28. What is a commonly seen type of dynamic route?
- Default static route
 - OSPF route
 - Link local route
 - DHCP route
29. If an interface goes down and it has a static route associated with it, what happens to that route?
- The static route is removed from the routing table
 - The router polls neighbors for a replacement route
 - The static route remains in the table because it was defined as static
 - The router automatically redirects the static route to use another interface
30. Which dynamic routing protocol was created to interconnect different ISPs?
- IS-IS
 - EIGRP
 - OSPF
 - BGP
31. Which routing protocol is used on larger network implementations because it simplifies route distribution but could be more complex to implement?
- EIGRP
 - RIP
 - OSPF
 - BGP
32. Why is the passive-interface command in OSPF used?
- Allows a router to send routing updates on an interface but not receive updates via that interface
 - Allows an interface to remain up without receiving keepalives
 - Allows interfaces to share IP addresses
 - Stops outgoing hello and update packets
33. Which level of the OSI 7-layer model is the Network Layer?
- 7th Level
 - 5th Level
 - 4th Level
 - 3rd Level

34. Which IP Address below is a valid private Class-A IP Address?
- a. 11.25.83.90
 - b. 10.12.255.12
 - c. 172.16.72.12
 - d. 192.168.95.30
35. Which IP Address below is a valid private Class-B IP Address?
- a. 172.16.57.2
 - b. 172.33.75.12
 - c. 11.0.86.44
 - d. 192.168.1.65
36. Which IP Address below is a valid private Class-C IP Address?
- a. 192.168.11.26
 - b. 192.168.1.257
 - c. 11.0.52.1
 - d. 172.16.13.4
37. Which protocol uses TCP?
- a. IPTV
 - b. TFTP
 - c. SMTP
 - d. DHCP
38. In what type of attack may these tools (nslookup and fping) be involved?
- a. Access attack
 - b. Reconnaissance attack
 - c. Denial of service attack
 - d. Worm attack
39. An administrator can reset a router's password without having physical access to it
- a. True
 - b. False
40. What is used to locate and load the IOS image?
- a. IOS image file
 - b. POST
 - c. BOOT environment variable
 - d. startup-config
41. What is a characteristic of a Layer 2 loop?
- a. A switch is constantly forwarding the same unicast frame
 - b. Routers are constantly forwarding packets
 - c. TTL on a packet is set to infinite
 - d. Broadcast frames are forwarded back to the original switch
42. What happens if there's a Layer 2 broadcast storm?
- a. Routers take control of the switching
 - b. New traffic is discarded by the switch since there is no way to process it
 - c. Switches become inaccessible through its console connection
 - d. New traffic will be forwarded but at lower speeds

43. What is the purpose of Spanning-Tree Protocol?
 - a. To prevent big broadcast domains
 - b. To prevent Layer 2 loops
 - c. To improve switch forwarding
 - d. To allow switches to route packets across networks
44. Which switching method send the frame as soon as it reads the destination MAC address?
 - a. Fragment free
 - b. FCS
 - c. Cut-through
 - d. Store-and-forward
45. Which RJ-45 cable is used to connect two switches together if no auto-negotiation protocols are being used?
 - a. Crossover Cable
 - b. Roll-over Cable
 - c. Straight Cable
 - d. Serial Cable
46. What happens to switch ports if you delete the VLAN that was assigned to it?
 - a. The ports are disabled
 - b. The ports are placed in trunk mode
 - c. The ports are assigned to VLAN1, the default VLAN
 - d. The ports stop communicating with the attached devices
47. Which mode is being used when the prompt shows "Switch(config-if)#"?
 - a. Interface Configuration
 - b. User Exec
 - c. Privilege Exec
 - d. Global Configuration
48. What does the reload command do on User EXEC mode?
 - a. Reboots the device without saving the configuration
 - b. Reboots the device while saving the configuration
 - c. Nothing happens
 - d. Rearms the security enabled on the ports of the device
49. A host receives a frame and discards it after determining it is corrupt. Which OSI layer does this?
 - a. Application
 - b. Physical
 - c. Network
 - d. Data-Link
50. What happens if a switch receives a frame with FF:FF:FF:FF:FF:FF as destination MAC address?
 - a. The switch will not forward the frame
 - b. The switch will send the frame to a connected router
 - c. The switch will forward the frame on all ports except the one it came in from
 - d. The switch will discard the frame